> **Prepared 2026-11-12 · developer**
>
> Development web source only. Module availability and hosted services require separate acceptance. Native and production deployment are separate.
>
> `Source 0a8824f4390bb89778c3a83e4c8f2c3da5257cf0`
>
> `Application 91f1d2c4bb2fe276527a7783e710b17da63495b240aeee1d208ae8651a55faf1`

NEVORA web developer guide

This distribution describes the accompanying development source. Its manifest binds source and guide bytes; it does not establish provider activation, hosted acceptance or production authorization. The broader developer reference supplies additional contract detail. Native apps and native push remain a separate track.

Establish your working source

Read AGENTS.md, the relevant repository skills, package.json, tests and the code you intend to change. Fetch and reconcile `origin/development` before new integration work. Inspect status and ownership; preserve other agents' changes and use a detached worktree for isolation. Only `development`, `main` and `claude-codex-collab-nevora` are active branches under the current policy. Main and production require Jose's explicit authorization.

Run `npm run check:development-sources` after fetching before assembly. Its inventory detects drift; it does not merge or approve sources. Record exact candidate identity, changed scope, checks actually run and remaining gates in the handoff.

Architecture and source ownership

The app uses Next.js App Router, React, TypeScript and PostgreSQL/Neon. The current company model has one company per database, a singleton workspace, locations and explicit employee eligibility. Isolated-company onboarding uses a separate control registry and guarded runtime bindings; its existence does not authorize changing an existing single-company deployment.

Layer	Responsibility and entry points
Protected shell and pages	`src/app/app/layout.tsx`, `src/app/app/**/page.tsx`, components and nearby server actions
Transport	`src/app/api/**/route.ts` and domain HTTP modules validate envelope, origin, size and inputs

Layer	Responsibility and entry points
Identity and authority	`src/lib/auth/`, `src/lib/company-access/`; obtain current server session/proof and current scope
Business rules and commands	Scheduling, shift changes, time corrections, payroll, notifications, onboarding and Owner-advisor modules under `src/lib/`
Persistence	`src/lib/data/`, guarded domain stores and `src/lib/db.ts`
Installation/runtime	`db/migrations/order.ts`, company setup and `src/lib/tenancy/`; use recorded migration/source contracts
Private operations	`src/lib/operations/`, `ops/workforce/`, canonical worker entry points

Trace a real command end to end before editing. A typical path is a form/server action or API route, parser/service, guarded transaction and bounded response or receipt. Presentation and pure validation do not establish database authority. Keep business decisions, transport, UI and data access separate. Claude owns product UI; coordinate a contract before changing that surface.

Preserve these contracts

- **Current authority:** `OWNER`, `MANAGER`, `LEAD` and `EMPLOYEE` are access roles, not positions. Manager operational scope is an Owner-assigned current company/location grant. Payroll grants and financial-object coverage are separate. Recheck authority at protected read/write boundaries and across awaited work; navigation filtering alone is insufficient.
- **Receipts and uncertainty:** retain the original operation ID and immutable command when recovering an uncertain result. A timeout can follow a committed write. Use the workflow's own recovery protocol; not every mutation implements the same receipt contract.
- **Reviewed versions:** preserve expected revisions, source fingerprints and review tokens. Do not refresh tokens behind the user to force a stale command through. Drafts, published schedule snapshots and acknowledgements are separate.
- **SQL versions:** preserve microsecond timestamp text under the owning transaction’s UTC, ISO/YMD and READ COMMITTED contract. Follow the upgrade compatibility procedure for Owner invitation reissue and fresh review. A merge performs no hosted migration.
- **Time and money:** calendar labels differ from instants and elapsed time. Use the workspace timezone and existing helpers. End-of-day `24:00` differs from the start-of-day `00:00`. Preserve exact payroll minor units, rounding and recorded-source rules.
- **Private reads:** ready-empty, unavailable, denied and expired are distinct. Private browser snapshots expire and clear on access/lifetime changes. Ignore late results from an older navigation, session or request.

- **Feature gates:** gate pages, actions and APIs as well as navigation. The communications D04 hold requires owned disposable local data; a feature flag is not a live activation bypass.
- **Rondo:** model tools read bounded evidence. Coverage mutations require the explicit Owner-reviewed command and commit-time checks. Saved goals are intent, not authority. Provider configuration does not prove provider authentication.

Onboarding and worker boundaries

Production admission defaults closed. The approved-recipient mode is a bounded inbox fence, not public signup. Mixed open admission plus a production inbox fence fails closed. Registration rejection precedes persistence. Provider creation requires explicit capacity policy, including the bounded total-project cap; pending/unknown attempts retain capacity until safely reconciled.

Keep the web, provisioner, enrolled worker and independent monitor principals separate. Do not expose privileged bootstrap or management credentials to the browser/web process. Fresh-company installation, existing-company upgrade and restore are different operations; never rerun the fresh seed chain on an existing company. See self-service operations and worker enrollment.

Verify safely

Inspect scripts before execution and use the version/locked dependencies in the current CI configuration. The declared `build` and `start` run Next.js directly; setup, workers and database harnesses have separate side effects. Read the engineering-toolkit README before using shared utilities.

For an ordinary code change, run the relevant focused regression first, make the smallest fix, and rerun it. The normal static/unit checks are:

```
npm test -- --maxWorkers=1
npm run typecheck
npm run lint
```

Use the specific owned PostgreSQL/browser runner for integration evidence. Read its ownership and cleanup controls first; use fictional, invocation-owned data. Never point tests at customer databases or use real SMTP/push as a fixture. A worker invocation can create jobs, documents or deliveries; it is not a harmless health probe. Do not run migrations by filename order.

Follow CI for source-bound builds and SQL/browser/package gates. Phase E requires six leaves and all 70 combinations: separate Home engines run seven widths each; other suites run both. Existing limits apply; missing, duplicate or mismatched evidence fails aggregation. Preserve failures, skips and independent findings. Record source SHA, environment, command, result and artifact identity; distinguish local, CI, development, hosted and production acceptance.

For guides, validate links/evidence, rebuild and verify PDFs with checked-in tooling. Never relabel older PDFs or hosted receipts. Keep secrets and private test content out of Git, logs and support records. See the operations guide for deployment custody and incident boundaries.